Lab01

#cyber-security

1

We can see and explore the Kali graphical environment, here we can see we've opened the home folder and viewing it's contents via the GUI tooling inside Kali.



We can find some of the tools Kali comes with via the dropdown menu. In the reconnaissance section we can find tools such as burpsuite, dirbuster, nmap, recon-ng and gobuster where

- burpsuite - A professional tool suite that includes paid packages. Meant for working with web applications and have tools such as interceptors and sniffers, repeaters, packet modifiers etc.
- dirbuster - A tool used for directory traversal within web applications to find any potentially sensitive information that has been left open
- gobuster - Does the same as dirbuster but also goes through other types of data such as AWS objects, FTP files, DNS records etc.
- nmap - Network reconnaissance tools with extensive features, enables scanning networks for online hosts, running services, OS and service versions etc.

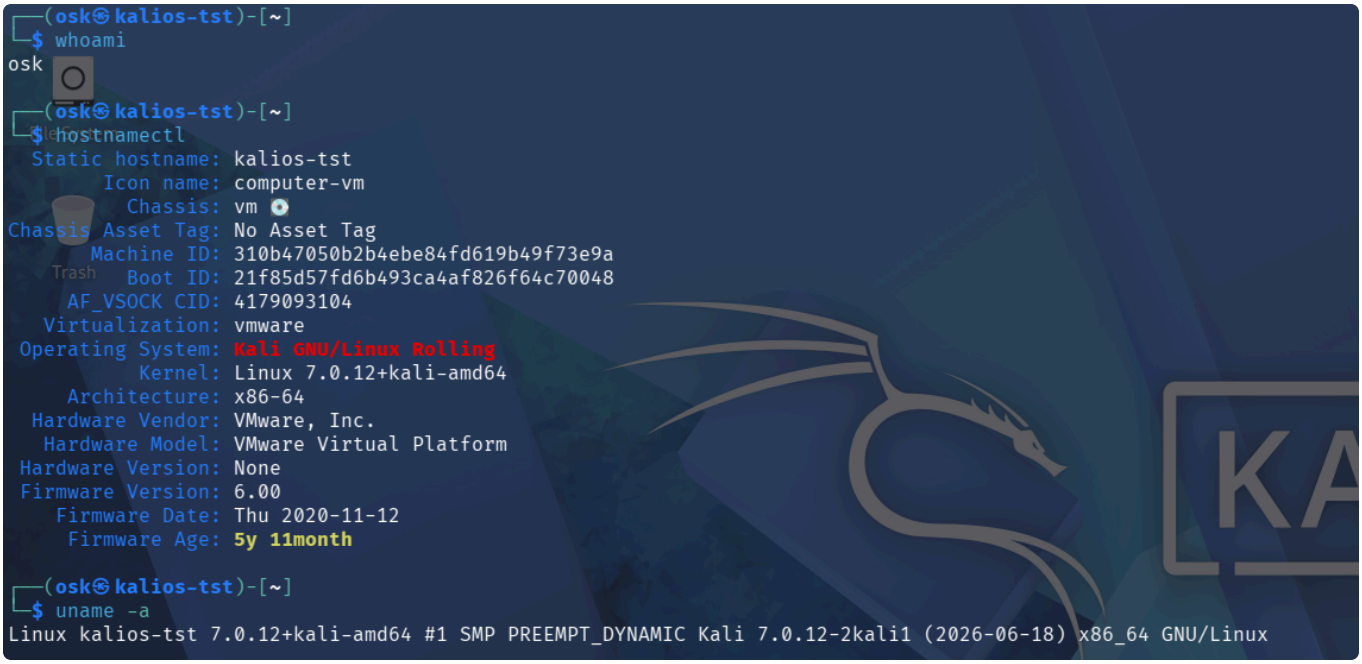
- recon-ng - OSINT reconnaisance tool.



2

We check the current user via `whoami`, get hostname information - information more particular to the machine, such as its name and OS via `hostnamectl` and then with `uname -a` we get

general system information.



Running `df -h` gets us information on the space we have available and the size of each filesystem.
Additionally we also check our current IP information via `ip a`, we can see that we have both an IPv4 as well as an IPv6 assigned to our device.



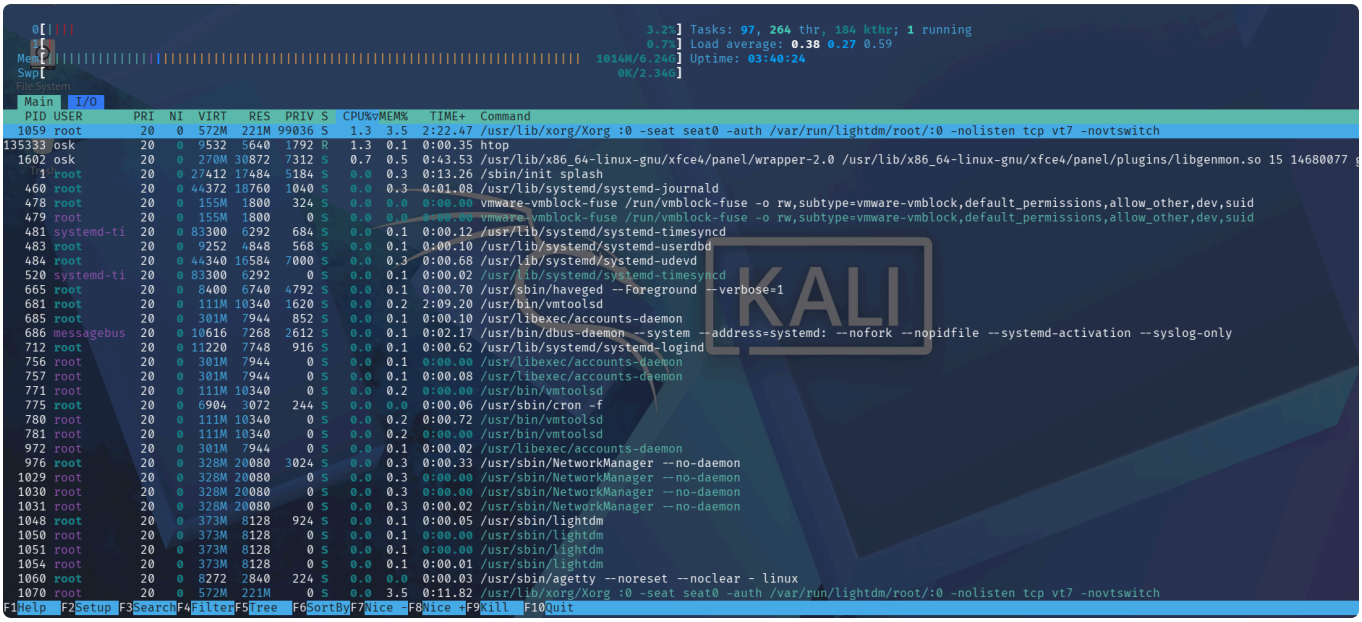
We use the `wget` command followed by a link to a resource on the internet to download it, here we download some image gotten from the web.

```
┌──(osk㉿kalios-tst)-[~]
└─$ wget https://i.natgeofe.com/n/8a4cd21e-3906-4c9d-8838-b13ef85f4b6e/tpc18-outdoor-gallery-1002418-12000351_01.jpg
--2026-10-13 12:13:00--  https://i.natgeofe.com/n/8a4cd21e-3906-4c9d-8838-b13ef85f4b6e/tpc18-outdoor-gallery-1002418-12000351_01.jpg
Resolving i.natgeofe.com (i.natgeofe.com)... 2.18.79.146, 2.18.79.133, 2a02:26f0:f700:14::210:109b, ...
Connecting to i.natgeofe.com (i.natgeofe.com)|2.18.79.146|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 955951 (934K) [image/jpeg]
Saving to: 'tpc18-outdoor-gallery-1002418-12000351_01.jpg'

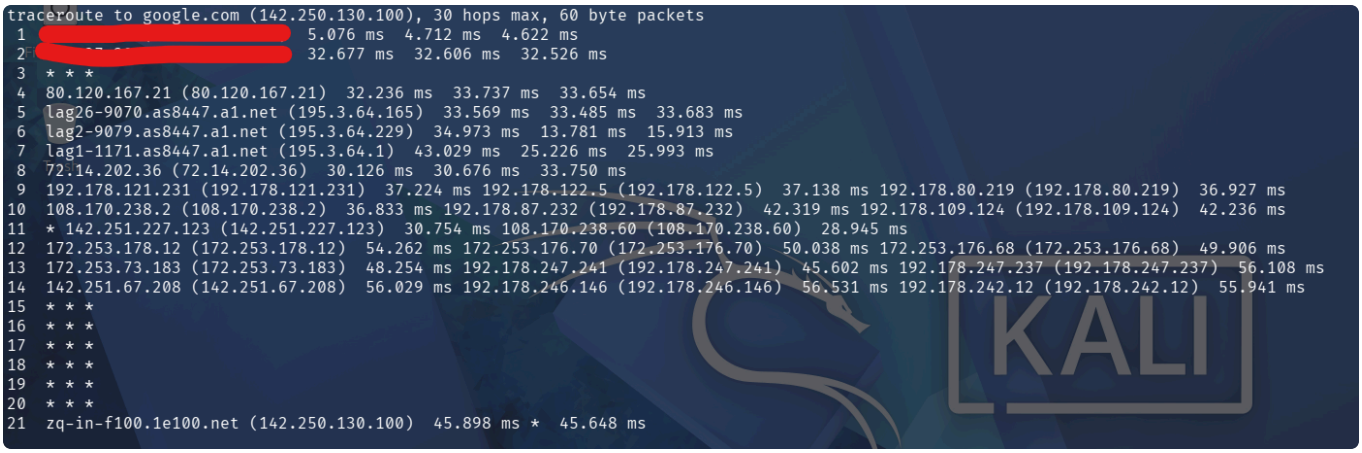
tpc18-outdoor-gallery-1002418-12000351_01.jpg   100%[===================>] 933.55K  5.43MB/s    in 0.2s

2026-10-13 12:13:00 (5.43 MB/s) - 'tpc18-outdoor-gallery-1002418-12000351_01.jpg' saved [955951/955951]
```

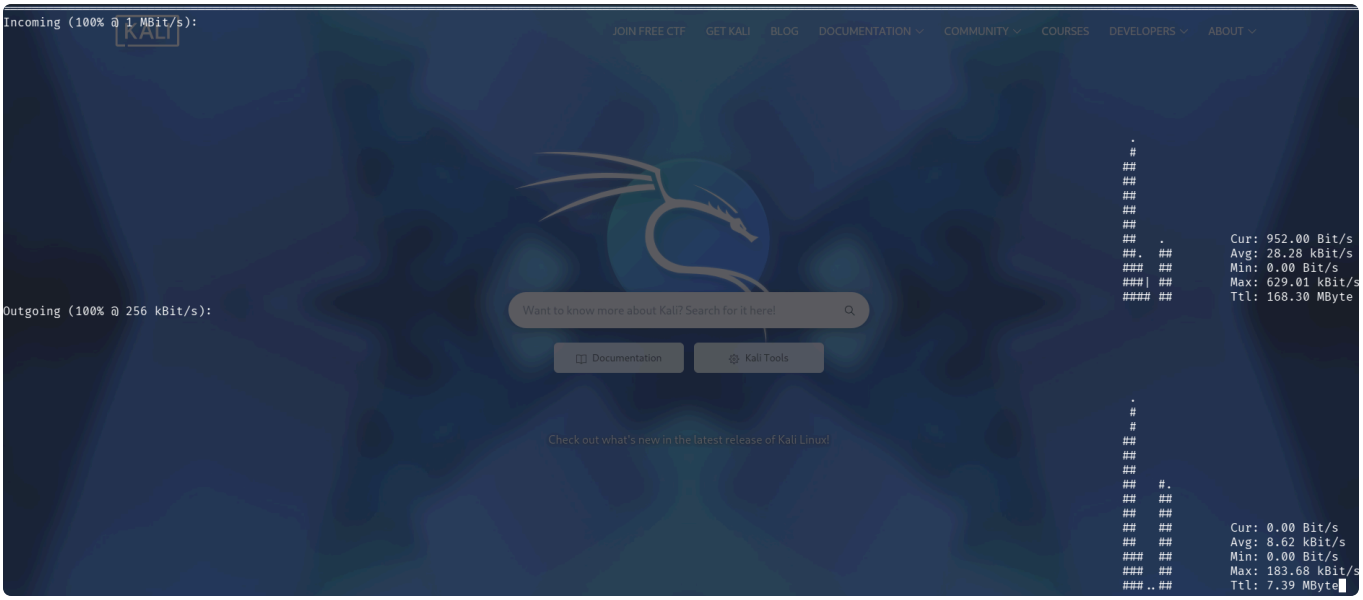
`htop` if installed, lets us view system resource usage in the terminal:



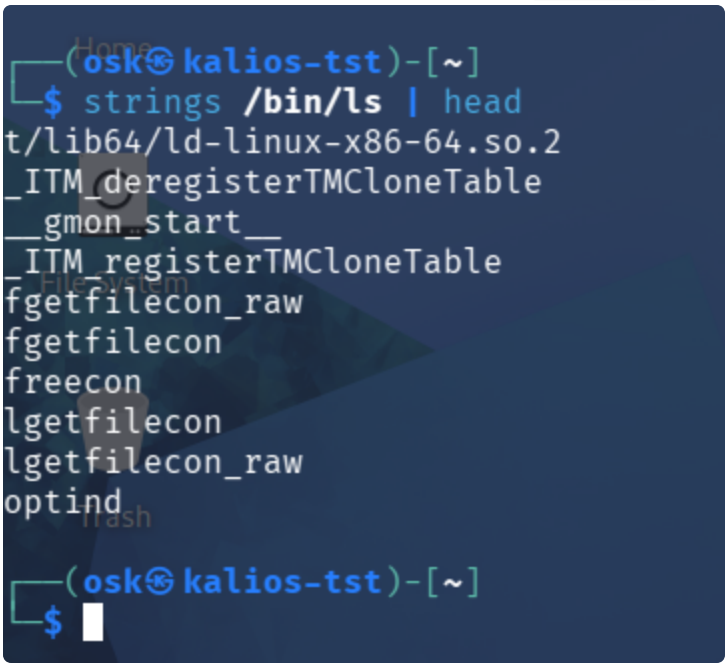
`traceroute` allows us to see the path a packet travels to reach a given destination address, here 'google.com'.



We can see network activity/traffic visualized with the package and command `nload`.



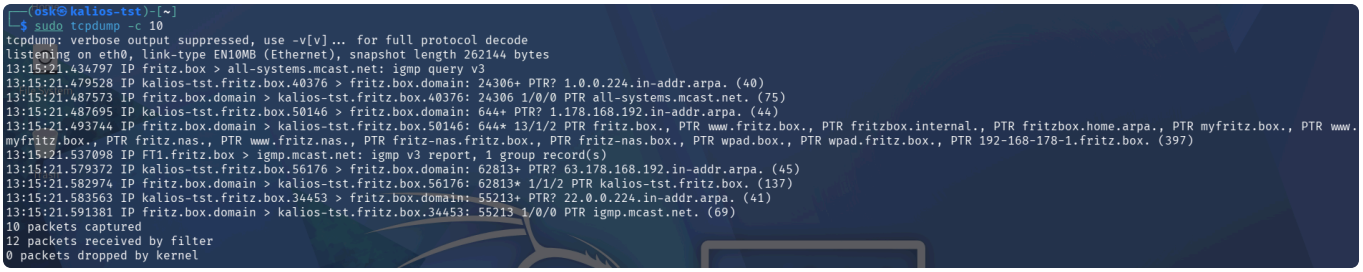
Binary data to string visualization via `strings` command.



3

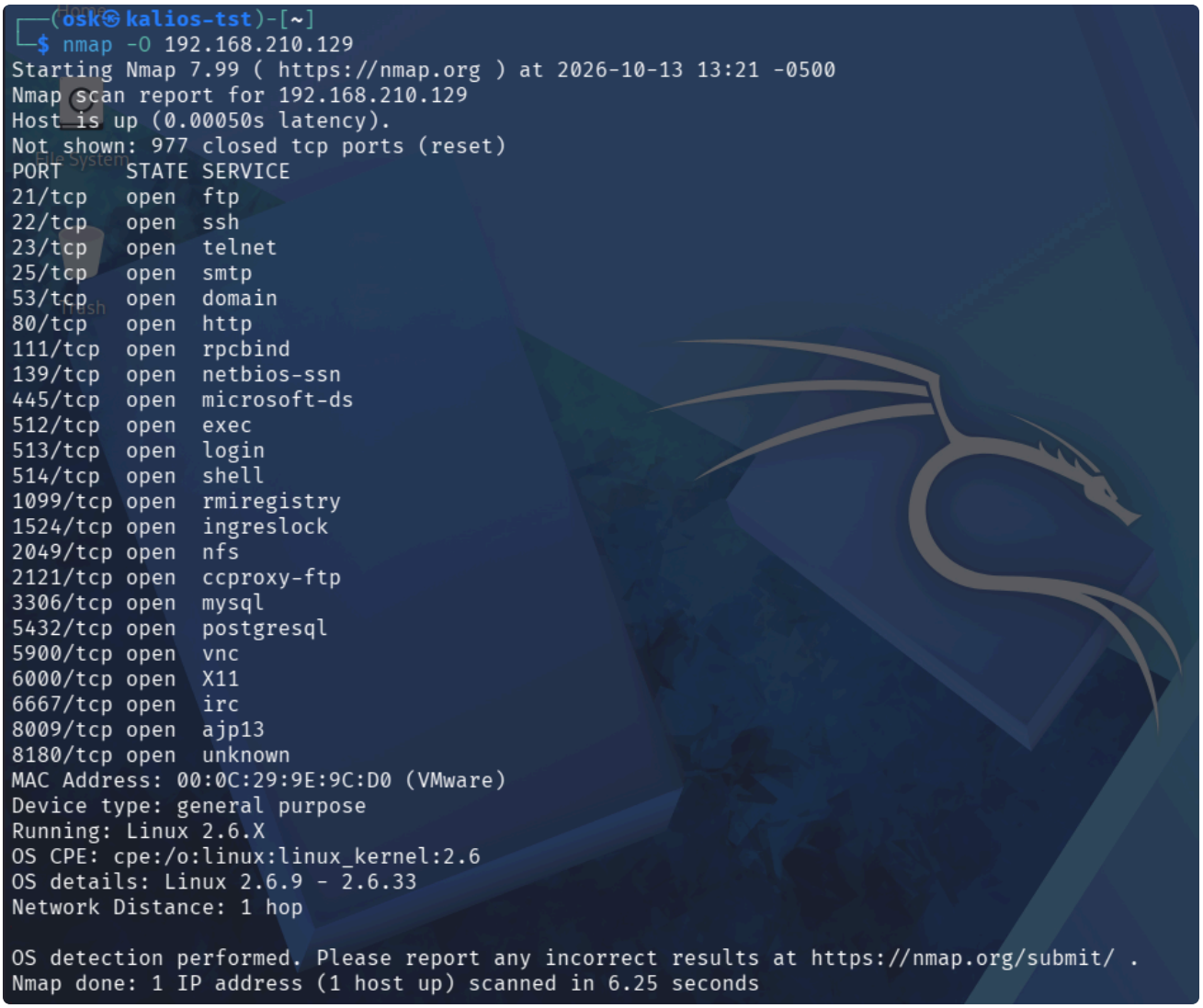
Speedtest-cli is not available for download on Kali.

We can dump the latest TCP packets that have travelled throughout the network via `tcpdump -c 10` - for the last 10 packets.



We can use nmap to scan for things such as services, ports, OS etc.
We get the following output when running `nmap -O 192.168.210.129`. Other parameters include but are not limited to:

- p - ports to scan
- T (0-5) - Setting how aggressive the scan will be, going from 0 (paranoid) to 5 (aggressive)
- sS - Setting the scan type to "stealth", meaning it will utilize TCP connection but never complete the handshake



We can use the `searchsploit` tool to search for publicly available vulnerabilities related to our search term.



4

We use Kali Linux typically for cybersecurity reasons, with that being so due to the fact that the distro, based on Debian, was made with the intention of being used for those purposes, with it coming packaged with a collection of different cybersecurity tools.
Regarding features the collection of tools is the most attractive attribute as it saves time and makes things more convenient especially for learning.